Tank Factors and Gas Consumption

A tank factor is the cubic feet of gas in every 100 psi of a cylinder. Know it and you can turn any pressure into volume, and any volume into pressure, in your head.

Tank factor = (rated capacity ÷ rated pressure) × 100 · Doubles = 2 × single

Quick reference, with consumption in doubles

Psi per minute in doubles at a SAC rate of 0.75 cu ft/min. Divide by your own tank factor for any other SAC rate.

Cylinder	Rated psi	Rated cu ft	Tank factor	Doubles	1 ATA 0 ft	2 ATA 33 ft	3 ATA 66 ft	4 ATA 99 ft	5 ATA 132 ft	6 ATA 165 ft	7 ATA 198 ft	8 ATA 231 ft
Steel, high pressure												
Faber FX-15	3,442	15	0.4	0.9	86	172	258	344	430	516	602	688
Faber FX-23	3,442	23	0.7	1.3	56	112	168	224	281	337	393	449
HP80	3,442	80	2.3	4.6	16	32	48	65	81	97	113	129
HP100	3,442	100	2.9	5.8	13	26	39	52	65	77	90	103
HP117	3,442	117	3.4	6.8	11	22	33	44	55	66	77	88
HP120	3,442	120	3.5	7.0	11	22	32	43	54	65	75	86
HP133	3,442	133	3.9	7.7	10	19	29	39	49	58	68	78
Steel, low pressure												
LP27	2,640	27	1.0	2.0	37	73	110	147	183	220	257	293
LP50	2,640	50	1.9	3.8	20	40	59	79	99	119	139	158
LP85	2,640	85	3.2	6.4	12	23	35	47	58	70	82	93
LP95	2,640	95	3.6	7.2	10	21	31	42	52	63	73	83
LP104	2,640	104	3.9	7.9	10	19	29	38	48	57	67	76
LP108	2,640	108	4.1	8.2	9	18	28	37	46	55	64	73

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Cylinder	Rated psi	Rated cu ft	Tank factor	Doubles	1 ATA 0 ft	2 ATA 33 ft	3 ATA 66 ft	4 ATA 99 ft	5 ATA 132 ft	6 ATA 165 ft	7 ATA 198 ft	8 ATA 231 ft
Aluminum												
AL6	3,000	6	0.2	0.4	—	—	—	—	—	—	—	—
AL13	3,000	13	0.4	0.9	—	—	—	—	—	—	—	—
AL40	3,000	40	1.3	2.7	28	56	84	112	141	169	197	225
AL80	3,000	77.4	2.6	5.2	15	29	44	58	73	87	102	116

AL80 uses its true capacity of 77.4 cu ft, so its factor is 2.6; UTD rounds it down to 2.5 for planning. Many low-pressure steel tanks reach their nominal capacity only with an overfill, so check your tank's spec sheet.

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Steel cylinder specifications

Cylinder	Service pressure (psi)	Capacity (cu ft)	Water volume (L)	Tank factor	Doubles tank factor	Diameter (in)	Length (in)	Empty weight (lb)	Buoyancy empty (lb)	Buoyancy full (lb)
OMS C8	2,640	7.87	1.2	0.3	0.6	4	11.25	5	-1.61	-2.23
OMS C13	2,640	13	2	0.5	1.0	4	14	5.9	-2.25	-3.31
Faber FX15DV	3,442	15	2	0.4	0.9	4	14.17	6.6	-1.21	-2.42
OMS C20	2,640	20	3	0.8	1.5	4	19.5	7.5	-1.5	-3
Faber FX23DV	3,442	23	3	0.7	1.3	4	19.5	8.6	-0.66	-2.64
Faber L27DV	2,640	25	4	0.9	1.9	5	14.43	11.7	-1.07	-2.95
Worthington LP27	2,640	27	4.3	1.0	2.0	6	14.5	11.9	-1	-3
PST LP45	2,640	45	7.16	1.7	3.4	6	23	19.6	-0.5	-3.7
OMS C45	2,640	46	7	1.7	3.5	6	23	17.6	0	-4
Faber L50DVB	2,640	48.7	8	1.8	3.7	6	25.2	18.9	+1.24	-2.43
OMS C50	2,640	50	7.8	1.9	3.8	6	25.2	19.92	-1.5	-4.8
PST E7-65	3,442	65	8.28	1.9	3.8	7	16.75	23.2	-1.5	-6.4
OMS C66	2,640	66	10	2.5	5.0	7	21	25	+1.67	-5.15
Worthington X7-65	3,442	66.4	8.2	1.9	3.9	7	16.7	25.1	-3.9	-8.7
Faber FX71DVB	3,442	71	9	2.1	4.1	7	21.26	24.7	-1.4	-6.97
Faber M71DVB	3,300	71	9	2.2	4.3	7	20.47	28.7	+0.9	-4.35
PST MP72	3,300	71.2	9.42	2.2	4.3	7	20.75	30	-6	-11.4
Worthington LP77	2,640	77	12	2.9	5.8	7	23.2	32.5	-1	-6.8
PST E7-80	3,442	80	10.19	2.3	4.6	7	19.75	26	-3.3	-9.3
Faber HP80-HDG	3,442	80	10.2	2.3	4.6	7	20.8	28.3	-1.74	-8.05
PST LP80	2,640	80.6	12.86	3.1	6.1	7	24	34.4	-1	-7
Worthington X7-80	3,442	81	10.1	2.4	4.7	7	19.8	29.9	-3	-9
Faber L85DVB	2,640	81.1	13	3.1	6.1	7	25.98	31.2	+2.32	-3.8
Worthington LP85	2,640	82.9	12.8	3.1	6.3	7.25	24.7	36.8	-0.7	-7.1
OMS C85	2,640	85	13	3.2	6.4	7	26	31	0	-6.7
Worthington LP95	2,640	93.3	14.8	3.5	7.1	8	23.7	41.9	-3	-10.1
Faber L95DVB	2,640	95	15	3.6	7.2	8.02	23.82	37.2	+1.69	-5.37
OMS C98	2,640	98	15	3.7	7.4	8	24	38	0	-7.73
Worthington X7-100	3,442	99.5	12.2	2.9	5.8	7.25	22.7	33.1	-2.5	-10
Faber M100DVB	3,498	100	12	2.9	5.7	7.24	24.21	38.7	-6.69	-14.11
PST E7-100	3,442	100	12.73	2.9	5.8	7.25	24	32	-1.3	-8.8
Faber HP100-HDG	3,442	100	12.9	2.9	5.8	7.25	25.3	34	-0.59	-8.41

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Cylinder	Service pressure (psi)	Capacity (cu ft)	Water volume (L)	Tank factor	Doubles tank factor	Diameter (in)	Length (in)	Empty weight (lb)	Buoyancy empty (lb)	Buoyancy full (lb)
Faber L108DVB	2,640	108	17	4.1	8.2	8.02	26.77	41	+2.98	-5.02
Worthington LP108	2,640	108.6	16.8	4.1	8.2	8	26.5	45.95	-2.6	-10.7
OMS C108 (112)	2,640	112	17	4.2	8.5	8	26	41	-1	-8
Faber HP117-HDG	3,442	117	15	3.4	6.8	8	24.2	37.6	+0.35	-9.12
PST E8-119	3,442	119	15.17	3.5	6.9	8	24	39.3	-3.3	-12.2
PST E7-120	3,442	120	15.29	3.5	7.0	7.25	28	36.6	-1.3	-10.3
Faber HP120-HDG	3,442	120	15.3	3.5	7.0	7.25	29.1	39.2	+0.65	-8.82
Worthington LP121	2,640	120	18.9	4.5	9.1	8	29.2	49.9	-2	-11.1
Faber L120DVB	2,640	120	19	4.5	9.1	8.02	29.33	44.9	+4.07	-4.87
Worthington X7-120	3,442	120.6	15.3	3.5	7.0	7.25	27.7	39.7	-2	-11
Worthington X8-119	3,442	123	14.8	3.6	7.1	8	23.9	42.5	-2	-10.9
OMS C121	2,640	125	19	4.7	9.5	8	29	45	0	-9.5
PST E8-130	3,442	130	16.6	3.8	7.6	8	25.56	42.2	-4.8	-14.6
OMS C135	2,640	131	20	5.0	9.9	8	30.7	47	+0.75	-10.3
Worthington X8-130	3,442	131.4	16	3.8	7.6	8	25.3	44.7	-2	-11.7
Faber HP133-HDG	3,442	133	17	3.9	7.7	8	26.85	42.4	+1.45	-9.08
PST E8-149	3,442	149	18.98	4.3	8.7	8	29.37	47.5	-1.7	-12.8
Faber FX149DVB	3,442	149	19	4.3	8.7	8.02	29.53	46.9	+2.35	-9.41

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Aluminum cylinder specifications

Cylinder	Service pressure (psi)	Capacity (cu ft)	Water volume (L)	Tank factor	Doubles tank factor	Diameter (in)	Length (in)	Empty weight (lb)	Buoyancy empty (lb)	Buoyancy full (lb)
Catalina S4	3,000	4	0.57	0.1	0.3	3	7.8	1.9	−1.4	−1.6
Catalina S6	3,000	6	0.87	0.2	0.4	3	10.8	2.6	−1.2	−1.5
Luxfer S006	3,000	6.1	0.88	0.2	0.4	3	10.87	2.72	−1.03	−1.49
Metal Impact S06	3,000	6.1	0.9	0.2	0.4	3	11.1	2.5	−1	−1.4
Metal Impact S08	2,015	7.6	1.5	0.4	0.8	4	10.8	3.8	+2	+1.5
Catalina S13	3,000	13	1.87	0.4	0.9	4	12.8	5.7	−0.9	−1.8
Metal Impact S13	3,000	13.2	1.9	0.4	0.9	4	13.1	6	−0.9	−1.8
Luxfer S13S	3,000	13.2	1.9	0.4	0.9	4	12.87	5.94	−0.72	−1.71
Luxfer S014	2,015	13.7	2.82	0.7	1.4	4	16.2	4.9	+1.8	+0.8
Catalina S19	3,000	19	2.72	0.6	1.3	4	17.7	7.8	−0.2	−1.5
Luxfer S019	3,000	19.9	2.87	0.7	1.3	4	18.56	8.09	+0.12	−1.37
Metal Impact S19	3,000	19.9	2.9	0.7	1.3	4	18.6	8	−0.1	−1.3
Catalina S20	3,000	20	2.88	0.7	1.3	5	13.8	9.2	0	−1.4
Metal Impact S30	3,000	30	4.3	1.0	2.0	5	19.3	12.25	0	−1.5
Catalina S30	3,000	30	4.31	1.0	2.0	5	19.9	13.6	−0.2	−2.4
Luxfer S030	3,000	30	4.31	1.0	2.0	5	21.85	11.61	+1.2	−1
Metal Impact S40	3,000	40	5.7	1.3	2.7	5	24.6	15.3	+2.5	−0.7
Catalina S40	3,000	40	5.74	1.3	2.7	5	25.4	16.2	+1.4	−1.5
Luxfer S040	3,000	40	5.74	1.3	2.7	5	24.6	15.3	+1.8	−0.7
Catalina S45	3,000	45	6.46	1.5	3.0	7	17.7	20.3	+0.5	−2.8
Metal Impact S50	3,000	48.4	6.9	1.6	3.2	7	19	21.3	+0.8	−2.5
Luxfer S050	3,000	48.4	6.95	1.6	3.2	7	19	21.2	+1.3	−2.4
Catalina S50	3,000	50	7.18	1.7	3.3	7	19.4	21.8	+1.2	−2.4
Catalina S53	3,000	53	7.6	1.8	3.5	7	19.2	25.1	+0.1	−3.8
Catalina C60	3,300	60	7.96	1.8	3.6	7	19.9	27.3	−1.2	−5.6
Metal Impact S63	3,000	63	9	2.1	4.2	7	21.9	26.9	+1.7	−2.4
Catalina S63	3,000	63	9.05	2.1	4.2	7	21.7	26.9	+1.8	−2.8
Luxfer S063	3,000	63	9.05	2.1	4.2	7	21.85	26.6	+2.6	−2.1
Luxfer S072	3,000	69.6	10	2.3	4.6	7	26	28.5	+3.8	−1.4
Metal Impact AL72	3,000	72.1	10	2.4	4.8	7	26	27.8	+2.8	−0.7
Catalina C80	3,300	77.4	10.24	2.3	4.7	7	25.5	34.7	0	−5.8
Luxfer S080N	3,300	77.4	10.32	2.3	4.7	8	22.93	35.12	+2.26	−3.6

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Cylinder	Service pressure (psi)	Capacity (cu ft)	Water volume (L)	Tank factor	Doubles tank factor	Diameter (in)	Length (in)	Empty weight (lb)	Buoyancy empty (lb)	Buoyancy full (lb)
Metal Impact S80	3,000	77.4	11.1	2.6	5.2	7	26.1	31.9	+3.3	-1.5
Luxfer S080	3,000	77.4	11.11	2.6	5.2	7	26.06	31.38	+4.4	-1.4
Catalina S80	3,000	77.4	11.11	2.6	5.2	7	25.9	31.3	+4.1	-1.6
Metal Impact S100	3,300	98.8	12.2	3.0	6.0	8	26.2	41.1	-1.9	-4.6
Luxfer S100	3,300	98.8	13.18	3.0	6.0	8	26.2	41.1	+2.8	-4.6
Catalina C100	3,300	100	13.22	3.0	6.1	8	26.5	42	+2.2	-5.2

Compiled from manufacturer specification sheets. OMS weights include the valve; buoyancy figures include the valve. For Faber, hot-dip galvanized (HDG) models are listed where they overlap with painted versions. Always confirm against your own cylinder's markings and spec sheet.

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